

Exascale Computing Research OB1 cluster access documentation

This document describes the different steps you need to follow to access the **OB1** cluster.

READ THIS DOCUMENT FULLY AND THOROUGHLY BEFORE ATTEMPTING THE PROCEDURES DESCRIBED BELOW

0) Generating authentication RSA key pairs:

If you do not already have ssh keys generated on your workstation, you should generate a pair using the following command:

```
user@laptop$ ssh-keygen
```

This command will create an RSA key pair in `~/.ssh/id_rsa.pub` (**public** key) and `~/.ssh/id_rsa` (**private** key) that will allow you to authenticate without using passwords. If you do not want to use RSA, you can use any other algorithm of your choice.

1) Accessing the frontal node:

You need to login to **fob1** (frontal OB1) first with the provided initial credentials to renew your password and copy your public ssh key. To do so, you need to add the following lines to your `~/.ssh/config` on your local workstation:

```
Host fob1
  Hostname 193.55.217.28
  User USERNAME
  IdentityFile PATH_TO_PRIVATE_KEY ← Optional
```

Make sure to replace **USERNAME** by the given username (**userID**). Then, you can connect using the following command:

```
user@laptop$ ssh fob1
```

You'll be asked to change your password at first. Once your password is changed successfully, you should copy your public ssh key into **fob1** using the following command:

```
user@laptop$ ssh-copy-id -i ~/.ssh/id_rsa.pub fob1
```

After authentication has succeed, you will be able to access **fob1** without having to enter any authentication tokens. The following command

```
user@laptop$ ssh fob1
```

should connect you to the **fob1** gateway node automatically.

2) Accessing the Haswell compute (hswX) nodes:

Four Haswell compute nodes are available: hsw01, hsw03, hsw05, and hsw06.

Once you are connected to **fob1**, you should create local ssh keys for connecting to the compute nodes:

```
userID@fob1$ ssh-keygen
```

Once the keys generated, you can initiate the password reset con the compute nodes by trying to connect to each one:

```
userID@fob1$ ssh hsw01
```

Once you reset your password, you can copy your public ssh key on the compute node hsw01:

```
userID@fob1$ ssh-copy-id -i ~/.ssh/id_rsa.pub hsw01
```

The key does not need to be copied on the other nodes but the password must be reset.

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